

中文摘要

近年來，非地球同步軌道（non-geostationary satellite orbit, NGSO）衛星系統，特別是低地球軌道（low Earth orbit, LEO）衛星星座，已成為未來寬頻衛星通訊與非地面網路的重要發展方向。然而，當 LEO 衛星下行傳輸與既有地球同步軌道（geostationary satellite orbit, GSO）系統共用相同或相鄰頻段時，可能對 GSO 地面站造成同頻干擾。為保護 GSO 系統，NGSO 系統必須遵守等效功率通量密度（equivalent power flux-density, EPFD）限制。當 LEO 衛星通過 GSO 地面站附近，且與 GSO 方向形成近似共線幾何時，部分高干擾風險波束可能需要被關閉以滿足 EPFD 限制。然而，波束關閉雖然能有效降低干擾，卻會使原本由關閉波束服務的使用者失去服務，進而影響 LEO 使用者的服務連續性與滿足率。

為解決上述問題，本文提出 Safe-Beam Assisted Power-Release Relay (SAPR-R)，一種適用於 NGSO-GSO 共存情境下多波束 LEO 衛星系統的分階段中繼式服務恢復方法。SAPR-R 利用 LEO 衛星軌道與波束覆蓋範圍可預測的特性，預先建立 critical satellite、helper satellites、closed beams、safe beams，以及 critical/helper 衛星之間的波束重疊對應關係。當 critical satellite 為滿足 EPFD 限制而關閉部分高風險波束時，SAPR-R 先透過 critical satellite 上仍可運作的 safe beams 接手部分 helper users，以釋放 helper satellite 的可用功率；接著將釋放出的功率分配給 helper relay beams，以提升其接手 closed-beam users 的能力；最後，透過 beam-pair local relay 將受 critical-beam shutdown 影響的使用者重新分配至可行的 helper relay beams，完成服務恢復。

本文透過 STK 與 MATLAB 建立模擬環境，評估 SAPR-R 在 critical satellite 通過 GSO 地面站附近期間的 EPFD 合規性與使用者服務表現。模擬結果顯示，SAPR-R 能使 aggregate EPFD 維持在規範限制以下，並相較於 baseline methods 達到更高的平均使用者滿足率。此外，SAPR-R 能提升成功被 helper relay beams 接手的 closed-beam users 數量，並降低低滿足率使用者的比例。整體而言，SAPR-R 能在滿足 EPFD 限制的同時，有效改善 LEO 使用者的服務連續性，並避免在每個 time slot 中求解高複雜度的全域最佳化問題，提供一種兼具實用性與可行性的中繼式服務恢復方法。

關鍵詞：NGSO-GSO 共存、EPFD、LEO 衛星、多波束衛星系統、
服務恢復、中繼、非地面網路、SAPR-R

Abstract

In recent years, non-geostationary satellite orbit (NGSO) systems, especially low Earth orbit (LEO) satellite constellations, have become an important enabler for future broadband satellite communications and non-terrestrial networks. However, when LEO downlink transmissions operate in the same or adjacent frequency bands as existing geostationary satellite orbit (GSO) systems, they may cause co-channel interference to GSO ground stations. To protect GSO systems, NGSO systems must comply with equivalent power flux-density (EPFD) limits. When an LEO satellite passes near a GSO ground station and forms a near-inline geometry with respect to the GSO direction, some high-risk beams may need to be shut down to satisfy the EPFD limit. Although beam shutdown can effectively reduce interference, it also interrupts the service of users originally served by the closed beams and degrades LEO user service continuity and satisfaction.

To address this issue, this thesis proposes Safe-Beam Assisted Power-Release Relay (SAPR-R), a staged relay-based service recovery method for multi-beam LEO satellite systems under NGSO–GSO coexistence. SAPR-R exploits the predictable motion of LEO satellites and beam footprints to pre-compute the critical satellite, helper satellites, closed beams, safe beams, and beam overlap mappings between critical and helper satellites. When the critical satellite shuts down high-risk beams to satisfy the EPFD limit, SAPR-R first lets safe beams on the critical satellite take over selected helper users to release available power on the helper satellite; then allocates the released power to helper relay beams to improve their capability to serve closed-beam users; finally, it reassociates affected users to feasible helper relay beams through beam-pair local relay to complete service recovery.

An STK–MATLAB simulation framework is developed to evaluate the EPFD compliance and user service performance of SAPR-R during the critical satellite pass near the GSO ground station. The results show that SAPR-R keeps the aggregate EPFD below the regulatory limit and achieves higher average user satisfaction than baseline methods. Moreover, SAPR-R increases the number of closed-beam users successfully served by helper relay beams

and reduces the proportion of low-satisfaction users. Overall, SAPR-R improves LEO user service continuity while maintaining EPFD compliance, without solving a high-complexity global optimization problem in every time slot, thereby providing a practical and feasible relay-based service recovery approach.

Keywords : NGSO–GSO coexistence, EPFD, LEO satellites, multi-beam satellite systems, service recovery, relay, non-terrestrial networks, SAPR-R.

致謝

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Chapter 1

Introduction

In recent years, non-geostationary satellite orbit (NGSO) systems, especially low Earth orbit (LEO) constellations, have become an important component of broadband satellite communications, non-terrestrial networks (NTN), and future 6G architectures [1–4]. Unlike geostationary satellite orbit (GSO) systems, which are located above the Earth’s equator and appear fixed with respect to the ground, LEO satellites move rapidly along their orbits and can provide lower latency and more flexible coverage. However, when NGSO downlink transmissions operate in the same or adjacent frequency bands as existing GSO systems, they may cause co-channel interference to GSO ground stations. In particular, when a LEO satellite passes close to the line-of-sight direction between a GSO ground station and its target GSO satellite, a near-inline geometry may occur, leading to a higher interference risk, as illustrated in Fig. 1.1.

To protect existing GSO systems from interference generated by NGSO systems, the International Telecommunication Union (ITU) specifies interference limits for NGSO systems in Article 22 of the Radio Regulations [5]. Among these limits, equivalent power flux-density (EPFD) is used to evaluate the aggregate interference generated by NGSO transmissions toward GSO systems. Since EPFD depends on satellite geometry, transmit power, antenna gain, propagation loss, and the off-axis angle observed at the GSO ground station, its value changes over time as LEO satellites move along their orbits. Therefore, an NGSO system must ensure that the aggregate EPFD received at the GSO ground station remains below the applicable EPFD limit.

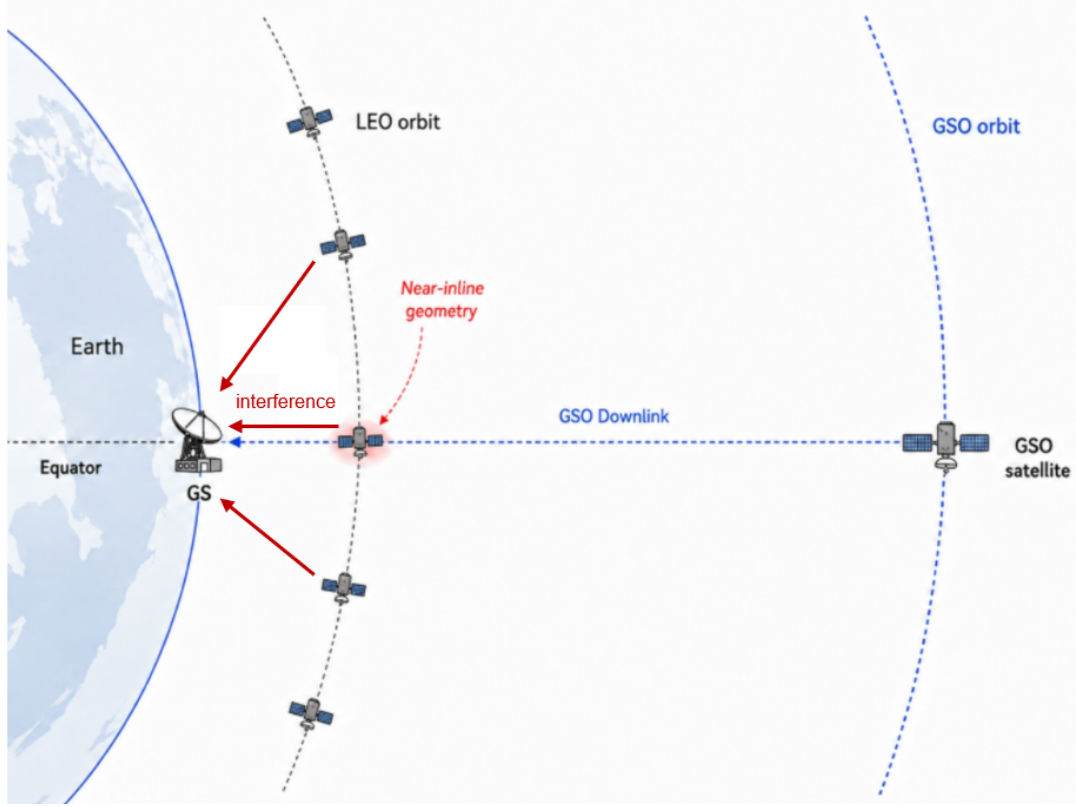


Figure 1.1: NGSO–GSO coexistence scenario and near-inline interference geometry between a LEO satellite, a GSO ground station, and the target GSO satellite.

To satisfy the EPFD constraint, NGSO systems may adopt different interference mitigation strategies, such as transmit power control, satellite tilt or pitch adjustment, beam shutdown, and beam steering [12–14]. These approaches can reduce the interference observed at the GSO ground station by lowering the transmitted power toward the GSO direction, changing the satellite pointing geometry, or avoiding high-risk beam transmissions. However, such mitigation actions may also reduce the available service resources for LEO users. For example, reducing beam power can decrease the achievable downlink rate, while shutting down high-risk beams can directly interrupt the service of users originally covered by those beams. Therefore, NGSO–GSO coexistence should not only be evaluated from the perspective of EPFD compliance, but also from the perspective of service continuity and user satisfaction.

In a multi-beam LEO system, the service impact of EPFD mitigation becomes more critical during near-inline events. When a LEO satellite passes close to the GSO ground station and the GSO line-of-sight direction, its beams may contribute a large portion of the

aggregate EPFD. To satisfy the EPFD limit, the system may need to shut down some high-risk beams on the dominant interfering satellite. Although beam shutdown is effective in reducing the EPFD contribution toward the GSO ground station, it also interrupts the service of users originally served by those closed beams. Therefore, the users under closed beams become the main service demand that must be recovered during the critical period.

This service recovery problem creates an opportunity for relay-based reassociation. LEO satellite motion and beam footprints are predictable, so the coverage relationship among neighboring satellites can be estimated in advance. In particular, same-orbit neighboring satellites may provide sufficient coverage overlap with the dominant interfering satellite during a critical period. Therefore, users affected by closed-beam shutdown may still be covered by neighboring LEO beams and can potentially be reassociated to these beams for service recovery. This observation motivates the use of neighboring satellites as helper satellites for relay-based user reassociation.

Based on this observation, this work proposes Safe-Beam Assisted Power-Release Relay (SAPR-R), a staged relay-based framework for maintaining user service under the EPFD constraint at the GSO ground station. The main idea is to exploit the predictable coverage overlap among neighboring LEO satellites. When high-risk beams are shut down to satisfy the EPFD limit, users originally served by those beams may still be covered by nearby helper beams. SAPR-R therefore uses relay-based user reassociation to transfer affected users from closed beams to feasible helper beams, so that their service can be continuously supported during the critical period. In this way, SAPR-R aims to improve user satisfaction while maintaining EPFD compliance.

The main contributions of this work are summarized as follows.

- We propose Safe-Beam Assisted Power-Release Relay (SAPR-R), a relay-based service recovery framework for EPFD-constrained multi-beam LEO systems. SAPR-R exploits predictable coverage overlap among neighboring LEO satellites to continuously support users affected by critical-beam shutdown.
- We design the core components of SAPR-R, including a time-slot beam relation record, safe-beam assisted power release, helper relay beam boosting, and beam-pair

local relay. The beam relation record captures the critical/helper satellite relationship and overlap-based beam mappings, while the staged relay procedure improves helper relay capability and reassociates closed-beam users to feasible helper beams.

- We evaluate SAPR-R using an STK–MATLAB simulation framework [20,21]. The results show that SAPR-R satisfies the EPFD constraint while improving average user satisfaction, increasing relay capability, and reducing the proportion of low-satisfaction users compared with baseline methods.

Chapter 2

Related Work

Existing studies on NGSO–GSO coexistence have investigated several interference mitigation strategies for reducing NGSO downlink interference toward GSO systems. In this chapter, related works are reviewed in three categories: progressive pitch and beam shutdown, joint power and tilt control, and beamforming-based interference mitigation. The characteristics and limitations of these methods are discussed to clarify the motivation of this work.

2.1 Progressive Pitch and Beam Shutdown

Progressive pitch is one representative attitude-based interference avoidance strategy for NGSO–GSO coexistence. In this approach, the LEO satellite gradually adjusts its attitude when it approaches low-latitude regions, where its downlink direction may become close to the line-of-sight direction from the GSO ground station to the target GSO satellite. By changing the satellite pointing direction, the LEO downlink beam can be shifted away from this line-of-sight direction, thereby increasing the angular separation observed at the GSO ground station. When attitude adjustment alone is insufficient to satisfy the interference constraint, some high-risk user beams may be shut down to further reduce interference.

Li et al. [14] studied the progressive pitch strategy of the OneWeb constellation. Their results show that progressive pitch can effectively reduce interference to GSO systems in many regions. However, the combination of satellite tilting and partial beam shutdown

may also reduce coverage continuity and create service blind areas during certain periods. This indicates that interference avoidance based only on attitude adjustment and beam shutdown may still affect the service availability of LEO users.

2.2 Joint Power and Tilt Control

Another type of interference mitigation method combines transmit power control with satellite tilt control. Power control reduces the transmit power of LEO satellites when they approach high-risk regions near the GSO ground station, so that the interference received at the GSO ground station can be lowered. Compared with progressive pitch methods that follow a predefined attitude adjustment strategy, joint power and tilt control formulates the mitigation process as an optimization problem. The transmit power and tilt angle of LEO satellites are jointly adjusted to satisfy the EPFD constraint while improving user demand satisfaction.

Jalali et al. [13] proposed a joint power and tilt control method for NGSO–GSO interference mitigation. Their method optimizes LEO transmit power and satellite tilt angle under the aggregate EPFD constraint. Compared with direct beam shutdown, this approach provides a more flexible way to reduce interference and preserve user service. However, when the interference constraint is strict, the transmit power of critical satellites may still need to be significantly reduced, which can limit the achievable service quality of LEO users during critical periods.

2.3 Beamforming and Beam Steering

Beamforming-based methods provide another interference mitigation direction for NGSO satellite systems equipped with phased-array antennas. Instead of only reducing transmit power or adjusting satellite attitude, beamforming controls antenna weight vectors to steer signals toward intended LEO users and suppress emissions toward GSO ground stations. In this way, the NGSO system can improve its downlink transmission while reducing co-channel interference to the GSO system.

Jalali et al. [15] investigated downlink beamforming strategies for interference-aware NGSO satellite systems. They proposed an aggregate interference-constrained beamforming method to limit the interference received at GSO ground stations while satisfying the service requirements of LEO users. This approach shows that beamforming can be an effective interference mitigation technique for advanced NGSO payloads. However, beamforming generally requires phased-array antennas, channel state information or average channel state information, and higher computational complexity. Since this work focuses on a fixed multi-beam LEO system, beamforming-based control is not considered in this thesis.

2.4 Summary and Research Gap

The above studies show that NGSO–GSO interference can be mitigated through satellite attitude adjustment, transmit power control, beam shutdown, or antenna-domain beamforming. These methods reduce the interference received at GSO ground stations from different perspectives. However, when mitigation actions reduce the available resources of LEO satellites, the service quality of LEO users may still be affected. In particular, users originally served by beams that are shut down during critical periods require additional service recovery support.

Existing studies have not addressed how neighboring LEO satellites can be used to recover the service of users affected by critical-beam shutdown under the GS EPFD constraint. This motivates the proposed SAPR-R framework, which exploits predictable coverage overlap among neighboring LEO satellites for relay-based service recovery while maintaining EPFD compliance.

Chapter 3

System Model and Problem

Formulation

3.1 System Model

1) Network Scenario: We consider a general multi-beam LEO satellite communication system under NGSO–GSO coexistence. Since the LEO downlink and the GSO system may operate in the same frequency band, the GSO ground station must be protected from excessive co-channel interference. Hereafter, this ground station is referred to as the GS. In the adopted simulation scenario, the GS is placed at the same longitude as the desired GSO satellite. This setting makes the GSO direction observed by the GS clear and allows LEO satellites passing near this direction to form a representative near-inline NGSO–GSO geometry. A OneWeb-like constellation with 16 fixed spot beams per satellite, as shown in Fig. 3.1, is used as a representative simulation setting, but the proposed framework can be applied to other LEO systems with predictable beam footprints and coverage overlap.

Let $\mathcal{S}(t)$ denote the set of visible LEO satellites at time slot t , and let $\mathcal{B}_s$ denote the set of beams on satellite s . In the adopted simulation setting, each satellite has 16 beams, while the formulation can be extended to systems with different beam numbers.

Each satellite employs a fixed multi-beam layout. For beam b on satellite s , the ground footprint at time slot t is denoted by $\mathcal{F}_{s,b}(t)$. If user u lies in $\mathcal{F}_{s,b}(t)$, beam b on satellite s is

regarded as a feasible serving candidate for that user. When multiple satellite–beam pairs cover the same ground area, the overlap provides possible serving alternatives through user association and resource coordination.

Based on the above GS–GSO geometry, a high-EPFD event may occur when a visible LEO satellite not only covers the GS vicinity but also appears close to the GS–GSO line-of-sight direction. The footprint coverage indicates that the LEO satellite is geographically close to the GS, while the small angular separation from the desired GSO satellite direction observed at the GS leads to a larger receive antenna gain toward the LEO direction. Therefore, its downlink transmission may produce a high aggregate EPFD at the GS. For each time slot t , the visible LEO satellite that contributes the largest EPFD risk at the GS is defined as the critical satellite, denoted by $s^C(t)$.

We further consider the along-track north and south neighboring satellites of the critical satellite as candidate helper satellites for subsequent service transfer and relaying, denoted by $s^N(t)$ and $s^S(t)$, respectively, as shown in Fig. 3.2. Since LEO satellite motion and beam footprints are predictable, the coverage overlap between critical beams and helper beams can be estimated in advance, enabling helper beams to relay users affected by critical-beam shutdown.

2) User Association and Service Model: Let $\mathcal{U}(t)$ denote the set of terrestrial users considered at time slot t , where each user is covered by at least one visible LEO beam. Each user can be associated with at most one satellite–beam pair in the same time slot. The user–beam association indicator $x_{u,s,b}(t)$ is defined as

$$x_{u,s,b}(t) = \begin{cases} 1, & \text{if user } u \text{ is served by beam } b \text{ of satellite } s, \\ 0, & \text{otherwise,} \end{cases}$$

for all $u \in \mathcal{U}(t)$, $s \in \mathcal{S}(t)$, and $b \in \mathcal{B}_s$.

For beam b on satellite s , the number of users served by this beam is

$$N_{s,b}(t) = \sum_{u \in \mathcal{U}(t)} x_{u,s,b}(t). \quad (3.1)$$

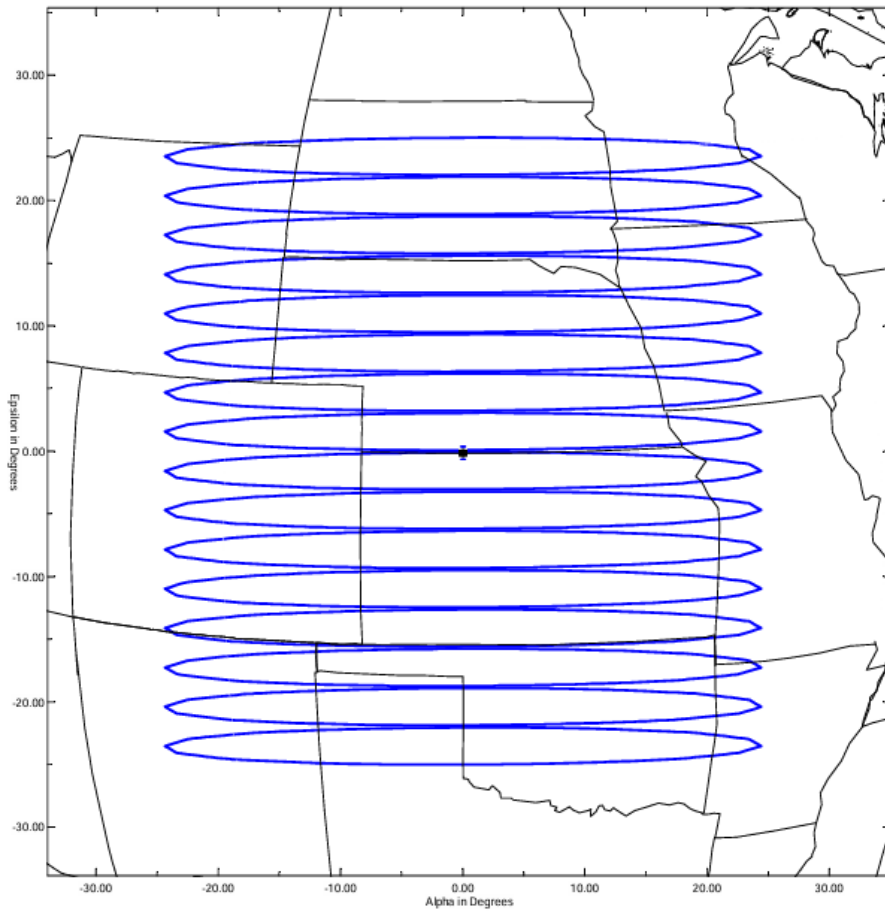


Figure 3.1: Ground footprints of the OneWeb Ku-band 16-beams

This value is used in the TDMA-based rate model, where users served by the same beam share the available downlink resource.

An association is feasible only when user u lies within the footprint $\mathcal{F}_{s,b}(t)$ of beam b on satellite s . Therefore, association, reassociation, and relay decisions are made only among satellite-beam pairs that can actually cover the user. In overlapping coverage areas, a user may be covered by multiple satellite-beam pairs. Under normal operation, these overlapping beams can coordinate the service demand through user association, time-domain scheduling, or spatial separation, while each user is still associated with only one serving beam in each time slot.

When a critical beam is shut down to satisfy the GS EPFD constraint, the users originally served by this beam can no longer be supported by the critical satellite. If these users are also covered by nearby helper beams, they can be reassociated to feasible helper relay beams. Since the helper relay beams use a compatible downlink service configuration,

such as the same service band and polarization setting, the reassociation can be described within the same service model.

3) Beam Power Model: We adopt a default beam-power operation model. For beam b on satellite s , the transmit power at time slot t is denoted by $P_{s,b}(t)$. Under normal operation, when LEO satellites are far from the GS and the EPFD violation risk is low, each active beam operates with its preconfigured default transmit power, denoted by $P_{s,b}^{\text{def}}$.

Different beams are not required to use the same default transmit power. The default power of each beam is preconfigured based on its link condition, so that users in different beam footprints can receive a similar service level under normal operation. Therefore,

$$P_{s,b}(t) = P_{s,b}^{\text{def}}, \quad \forall s \in \mathcal{S}(t), \forall b \in \mathcal{B}_s.$$

This default beam-power profile serves as the normal operating state. If the GS EPFD constraint cannot be satisfied under this state, some beam powers may be adjusted or shut down.

Each beam has a maximum transmit power $P_{s,b}^{\text{max}}$, and

$$0 \leq P_{s,b}(t) \leq P_{s,b}^{\text{max}}, \quad \forall s \in \mathcal{S}(t), \forall b \in \mathcal{B}_s. \quad (3.2)$$

Each satellite also has a total transmit power budget P_s^{max} :

$$\sum_{b \in \mathcal{B}_s} P_{s,b}(t) \leq P_s^{\text{max}}, \quad \forall s \in \mathcal{S}(t). \quad (3.3)$$

When the critical satellite $s^C(t)$ creates an aggregate EPFD violation risk at the GS, some critical beams may be shut down. Let $\mathcal{B}_C^{\text{off}}(t)$ denote the set of beams shut down on the critical satellite, also referred to as closed beams. Then

$$P_{s^C,b}(t) = 0, \quad \forall b \in \mathcal{B}_C^{\text{off}}(t).$$

The remaining active beams on the critical satellite are referred to as safe beams and

are denoted by $\mathcal{B}_C^{\text{safe}}(t)$, where

$$\mathcal{B}_C^{\text{safe}}(t) = \mathcal{B}_{s^c} \setminus \mathcal{B}_C^{\text{off}}(t).$$

These safe beams remain active with their default transmit powers:

$$P_{s^c,b}(t) = P_{s^c,b}^{\text{def}}, \quad \forall b \in \mathcal{B}_C^{\text{safe}}(t).$$

Hence, the transmit power of a beam on the critical satellite can be written as

$$P_{s^c,b}(t) = \begin{cases} 0, & b \in \mathcal{B}_C^{\text{off}}(t), \\ P_{s^c,b}^{\text{def}}, & b \in \mathcal{B}_C^{\text{safe}}(t). \end{cases}$$

For helper satellites, beams also operate with their default transmit powers under normal operation. If users affected by critical-beam shutdown are reassociated to feasible helper beams, the transmit power of those helper beams may be adjusted above their default values, subject to the beam-level maximum power and the total satellite power budget (3.3):

$$P_{s^h,b}(t) \leq P_{s^h,b}^{\text{max}}, \quad s^h \in \{s^N(t), s^S(t)\}.$$

Throughout this model, each user is still served by only one satellite–beam pair, and the service rate is determined by the selected serving beam and the users sharing that beam.

4) SINR Model: The downlink SINR of a user is modeled using a commonly adopted wireless and satellite communication formulation, where the received signal, co-channel interference, and receiver noise are considered [16]. Let $k_{s,b}$ denote the downlink channel used by beam b on satellite s . Only beams using the same downlink channel as the serving beam are included as co-channel interferers.

If user u is served by beam b on satellite s , its downlink SINR at time slot t is

$$\text{SINR}_u(t) = \frac{S_u(t)}{I_u^{\text{intra}}(t) + I_u^{\text{inter}}(t) + N_u(t)}. \quad (3.4)$$

where $S_u(t)$ is the desired signal power, $I_u^{\text{intra}}(t)$ is the intra-satellite co-channel interference, $I_u^{\text{inter}}(t)$ is the inter-satellite co-channel interference, and $N_u(t)$ is the receiver noise power.

Desired Signal Power: The desired signal power follows the standard satellite link-budget relation, and the propagation loss components are modeled according to ITU-R recommendations [9, 10]. For user u served by beam b on satellite s ,

$$S_u(t) = \frac{P_{s,b}(t) G_{s,b \rightarrow u}^{\text{tx}}(t) G_u^{\text{rx}}}{L_{s,u}(t)}.$$

where $P_{s,b}(t)$ is the transmit power of the serving beam, $G_{s,b \rightarrow u}^{\text{tx}}(t)$ is the transmit gain of beam b toward user u , G_u^{rx} is the user receive gain, and $L_{s,u}(t)$ is the total propagation loss between satellite s and user u .

The total propagation loss includes free-space path loss and atmospheric gaseous loss:

$$L_{s,u}(t) = L_{s,u}^{\text{fs}}(t) L_{s,u}^{\text{gas}}(t),$$

where $L_{s,u}^{\text{fs}}(t)$ is the free-space path loss and $L_{s,u}^{\text{gas}}(t)$ is the atmospheric gaseous loss.

The free-space path loss follows the standard free-space attenuation model in ITU-R P.525 [9]:

$$L_{s,u}^{\text{fs}}(t) = \left(\frac{4\pi d_{s,u}(t)}{\lambda} \right)^2,$$

where $d_{s,u}(t)$ is the distance between satellite s and user u , and λ is the carrier wavelength.

Intra-Satellite Interference: The intra-satellite interference comes from co-channel beams on the same satellite. Let $\mathcal{B}_s^{\text{co}}(b)$ denote the set of beams on satellite s that reuse the same channel as the serving beam b , excluding beam b itself. The intra-satellite interference is given by

$$I_u^{\text{intra}}(t) = \sum_{b' \in \mathcal{B}_s^{\text{co}}(b)} \frac{P_{s,b'}(t) G_{s,b' \rightarrow u}^{\text{tx}}(t) G_u^{\text{rx}}}{L_{s,u}(t)}.$$

Inter-Satellite Interference: The inter-satellite interference comes from co-channel beams on other visible satellites. Let $\mathcal{B}_{s'}^{\text{co}}(b)$ denote the set of beams on satellite s' that

reuse the same channel as the serving beam b : The inter-satellite interference is given by

$$I_u^{\text{inter}}(t) = \sum_{\substack{s' \in \mathcal{S}(t) \\ s' \neq s}} \sum_{b' \in \mathcal{B}_{s'}^{\text{co}}(b)} \frac{P_{s',b'}(t) G_{s',b' \rightarrow u}^{\text{tx}}(t) G_u^{\text{rx}}}{L_{s',u}(t)}.$$

where $G_{s',b' \rightarrow u}^{\text{tx}}(t)$ is the transmit gain of the interfering beam toward user u , and $L_{s',u}(t)$ is the total propagation loss between satellite s' and user u .

With this commonly used downlink SINR model, the user SINR changes when the serving beam, beam transmit power, propagation loss, or co-channel interference changes.

5) Capacity and User Satisfaction Model: After obtaining the user SINR, the instantaneous link capacity is modeled by the Shannon capacity formula, which is commonly used in wireless and satellite communication systems [17, 18]. Let B_u denote the effective bandwidth allocated to user u . The instantaneous capacity of user u at time slot t is

$$C_u(t) = B_u \log_2(1 + \text{SINR}_u(t)).$$

When multiple users are served by the same beam, they share the downlink resource of that beam. Following the time-division resource sharing model commonly used in satellite downlink systems [18, 19], the beam resource is assumed to be equally divided among the users served by the same beam. The actual service rate of user u is therefore

$$R_u(t) = \frac{C_u(t)}{N_{s,b}(t)}. \quad (3.5)$$

where $N_{s,b}(t)$ is defined in (3.1). This expression represents an equal-time sharing model, where users served by the same beam are allocated the same fraction of the beam resource in each time slot.

Let $D_u(t)$ denote the demand rate of user u at time slot t . The satisfaction of user u is defined as

$$\text{sat}_u(t) = \min\left(\frac{R_u(t)}{D_u(t)}, 1\right). \quad (3.6)$$

where $\text{sat}_u(t) \in [0, 1]$. If $\text{sat}_u(t) = 1$, the demand of user u is fully met; if $0 < \text{sat}_u(t) < 1$,

user u receives partial service; if $\text{sat}_u(t) = 0$, user u receives no effective service in the time slot.

This satisfaction metric represents the fraction of user demand that is satisfied in each time slot.

6) Beam-Level EPFD Model: The EPFD calculation follows the ITU Radio Regulations and ITU-R S.1503 framework for NGSO–GSO coexistence [5, 6]. Since different beams on the same LEO satellite have different pointing directions, antenna gains, and transmit powers toward the GS, the EPFD contribution is evaluated at the beam level and summed in the linear domain.

For EPFD calculation, let $P_{s,b}^{\text{ref}}(t)$ denote the radio-frequency (RF) transmit power of beam b on satellite s within the ITU reference bandwidth B_{ref} specified for the applicable EPFD limit [7, 8]. If $P_{s,b}(t)$ is defined over a wider beam bandwidth, it is converted to the corresponding power within B_{ref} before applying the EPFD formula. The beam-level EPFD contribution toward the GS is given by

$$\text{EPFD}_{s,b}(t) = \frac{P_{s,b}^{\text{ref}}(t) G_{s,b \rightarrow g}^{\text{tx}}(t)}{4\pi d_{s,g}^2(t)} \cdot \frac{G_g^{\text{rx}}(\theta_{g \rightarrow s}(t))}{G_{g,\text{max}}^{\text{rx}}},$$

where $G_{s,b \rightarrow g}^{\text{tx}}(t)$ is the transmit antenna gain of beam b on satellite s in the direction of the GS, $d_{s,g}(t)$ is the distance between satellite s and the GS, $G_g^{\text{rx}}(\theta_{g \rightarrow s}(t))$ is the receive antenna gain of the reference GS antenna in the direction of satellite s , $G_{g,\text{max}}^{\text{rx}}$ is the maximum receive antenna gain of the reference GS antenna, and $\theta_{g \rightarrow s}(t)$ is the off-axis angle between the GS antenna boresight and the direction of satellite s .

The transmit antenna gain $G_{s,b \rightarrow g}^{\text{tx}}(t)$ is evaluated from the beam antenna pattern according to the off-axis angle between the boresight of beam b and the direction of the GS. Since each beam has a different pointing direction, the EPFD contribution of each beam is computed individually.

The aggregate EPFD at the GS is obtained by summing the beam-level EPFD contri-

butions from all beams of all visible LEO satellites:

$$\text{EPFD}_{\text{agg}}(t) = \sum_{s \in \mathcal{S}(t)} \sum_{b \in \mathcal{B}_s} \text{EPFD}_{s,b}(t).$$

The resulting aggregate EPFD is compared with the applicable ITU EPFD limit in the problem formulation.

3.2 Problem Formulation

Based on the system model above, we formulate the user association and beam power control problem to maximize the total user satisfaction while satisfying association, power, and EPFD constraints. The optimization variables are the user–beam association indicators $\mathbf{x}(t) = \{x_{u,s,b}(t)\}$ and the beam transmit powers $\mathbf{P}(t) = \{P_{s,b}(t)\}$.

Objective function:

$$\max_{\mathbf{x}(t), \mathbf{P}(t)} \sum_{u \in \mathcal{U}(t)} \text{sat}_u(t). \quad (3.7)$$

Constraints:

$$\sum_{s \in \mathcal{S}(t)} \sum_{b \in \mathcal{B}_s} x_{u,s,b}(t) \leq 1, \quad \forall u \in \mathcal{U}(t), \quad (3.8)$$

$$\sum_{s \in \mathcal{S}(t)} \sum_{b \in \mathcal{B}_s} x_{u,s,b}(t) \geq 0, \quad \forall u \in \mathcal{U}(t), \quad (3.9)$$

$$0 \leq P_{s,b}(t) \leq P_{s,b}^{\max}, \quad \forall s \in \mathcal{S}(t), \forall b \in \mathcal{B}_s. \quad (3.10)$$

$$\sum_{b \in \mathcal{B}_s} P_{s,b}(t) \leq P_s^{\max}, \quad \forall s \in \mathcal{S}(t). \quad (3.11)$$

$$\text{EPFD}_{\text{agg}}(t) \leq \text{EPFD}_{\text{lim}}. \quad (3.12)$$

Constraint (3.8) ensures that each user is associated with at most one satellite–beam pair in each time slot.

Constraint (3.9) ensures that no user is assigned a negative beam association.

Constraint (3.10) limits the transmit power of each beam, including the possibility of beam shutdown when $P_{s,b}(t) = 0$.

Constraint (3.11) enforces the total transmit power budget of each satellite.

Constraint (3.12) ensures that the aggregate EPFD at the GS does not exceed the applicable ITU limit.

The objective in (3.7) maximizes the aggregate satisfaction of all users in the considered time slot. Although the objective is written as a summation of user satisfaction, each satisfaction value depends on the selected serving beam, beam loading, SINR, and service rate. In addition, beam power affects both the user-side downlink performance and the GS-side aggregate EPFD. Therefore, the association and power decisions are strongly coupled.

In critical GS geometries, reducing or shutting down beam power can lower the aggregate EPFD, but it may also reduce the service rate of users originally served by the affected beams. These users can be reassociated to feasible helper beams, but doing so changes the helper beam loading and may further affect the service rates of other users.

Because of these coupled association, power, SINR, service-rate, and EPFD relationships, the problem becomes a mixed-integer nonlinear optimization problem. Solving the global optimum in every time slot is computationally difficult. Therefore, the next chapter develops a staged heuristic framework to obtain feasible association and power decisions during critical periods.

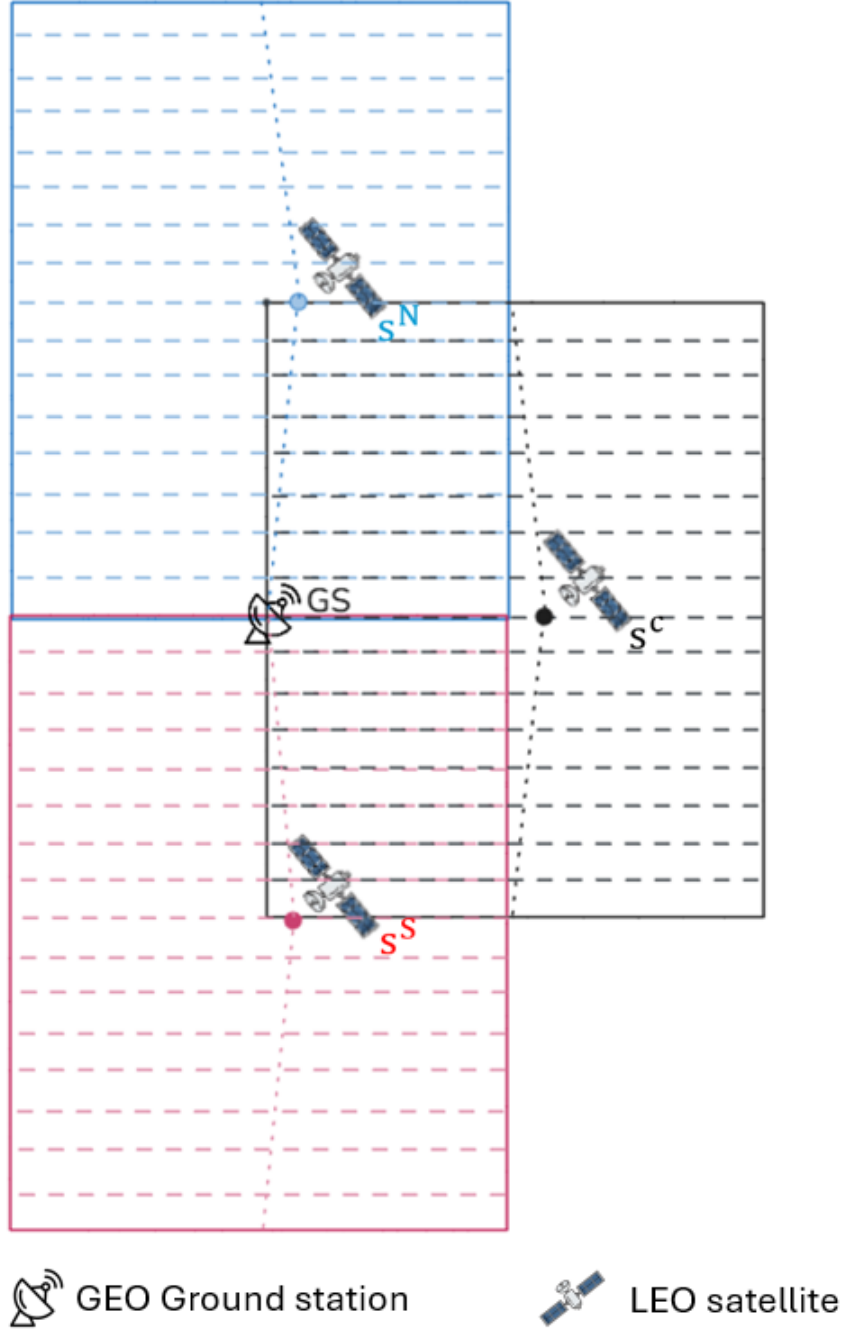


Figure 3.2: Network scenario with the GSO ground station GS , critical satellite s^C , and north/south helper satellites s^N and s^S .

Chapter 4

Method

4.1 Method Overview

Chapter III formulates the considered problem as a joint user association and beam power control problem under user association, beam power, satellite power budget, and EPFD constraints. The association variables are discrete, while the beam power variables are continuous, making the problem a mixed-integer nonlinear optimization problem. In addition, user association affects beam loading, downlink SINR, time-division resource sharing, and satisfaction, while beam power affects both user service quality and the aggregate EPFD at the GS. Directly solving the global optimum for every time slot is therefore computationally demanding.

To reduce online decision complexity, we propose Safe-Beam Assisted Power-Release Relay (SAPR-R). SAPR-R uses the predictable LEO satellite motion, GS location, GSO direction, and beam footprints to precompute beam-level relation records for all time slots in the observation window. For each time slot, the record includes the critical satellite, the north and south helper satellites, the closed beams on the critical satellite, the safe beams that remain active with default power, and the beam-pair mappings between critical beams and helper beams. This precomputation depends only on satellite geometry, beam footprints, and beam-level EPFD calculation, rather than user distribution, and can therefore be completed before online user reassociation.

During online operation, SAPR-R loads the beam relation record of the current time

slot and makes user-aware reassociation decisions based on user locations, current associations, and service satisfaction. First, Safe-Beam Assisted Helper Relief (SAHR) lets critical safe beams take over selected users originally served by helper release beams. A reassociation is accepted only when it does not reduce the satisfaction of the transferred user or the users already served by the critical safe beam. The transmit power no longer required by the helper release beams is then added to the available power budget of the helper satellite.

Next, the helper satellite allocates its available power budget to helper relay beams according to relay demand, so that these beams can provide additional service capacity for users affected by closed beams. Finally, Beam-Pair Local Relay (BPLR) reassociates closed-beam users to feasible helper relay beams within the precomputed beam-pair mappings. Candidate users are served sequentially according to relay link quality and remaining relay capacity.

In summary, SAPR-R consists of the following steps:

- **Time-slot beam relation record:** Precompute the critical/helper satellites, closed/safe beams, and beam-pair mappings for each time slot.
- **Safe-Beam Assisted Helper Relief (SAHR):** Use critical safe beams to take over selected helper users without reducing satisfaction, thereby releasing helper satellite power.
- **Helper relay beam boosting:** Allocate the available helper satellite power budget to helper relay beams according to relay demand.
- **Beam-Pair Local Relay (BPLR):** Reassociate closed-beam users to feasible helper relay beams within each beam pair according to relay link quality and remaining relay capacity.

4.2 Time-Slot Beam Relation Record

For each time slot t , SAPR-R constructs a beam relation record $\mathcal{R}_B(t)$, which stores the beam-level information used by the online reassociation and relay procedures. Figure 4.1 illustrates the time-varying beam status of the critical satellite and its helper satellites. The

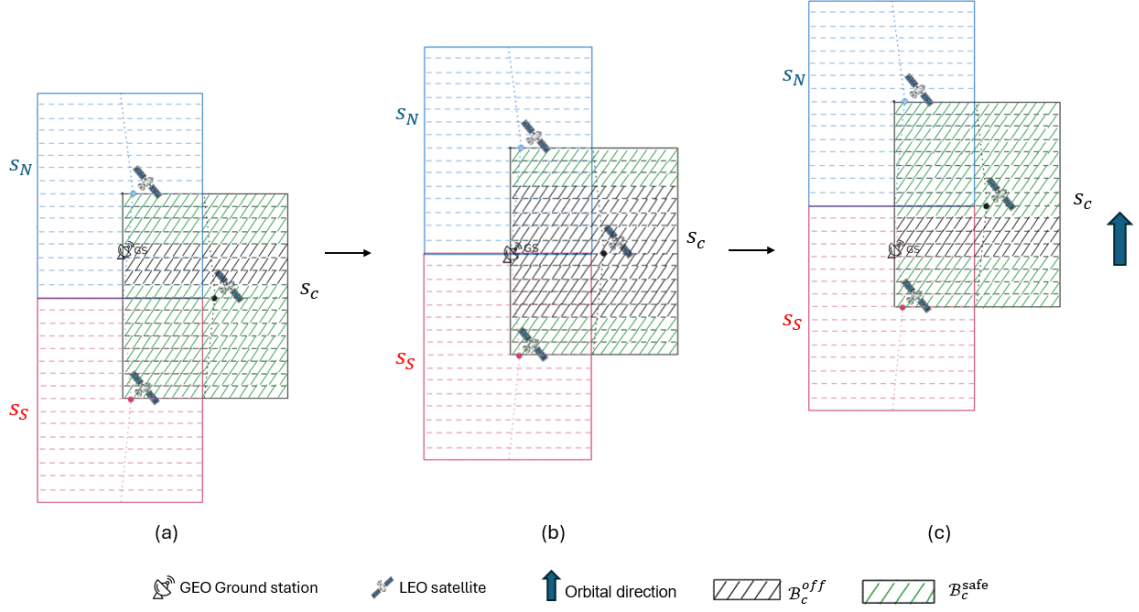


Figure 4.1: Three time slots as the along-track LEO satellites (s^N , s^c , and s^S) pass the protected GSO ground station. In region S_C on the critical satellite, black-hatched beams are $\mathcal{B}_c^{\text{off}}(t)$ (closed beams) and green-hatched beams are $\mathcal{B}_c^{\text{safe}}(t)$ (safe beams). Panels (a)–(c) show earlier to later time slots.

record is defined as

$$\begin{aligned} \mathcal{R}_B(t) = & (s^c(t), s^N(t), s^S(t), \\ & \mathcal{B}_c^{\text{off}}(t), \mathcal{B}_c^{\text{safe}}(t), \\ & \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t), \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)). \end{aligned}$$

In this record, $s^c(t)$, $s^N(t)$, and $s^S(t)$ denote the critical satellite, north helper satellite, and south helper satellite, respectively. $\mathcal{B}_c^{\text{off}}(t)$ denotes the set of closed beams on the critical satellite, and $\mathcal{B}_c^{\text{safe}}(t)$ denotes the set of safe beams that remain active with default power. $\mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$ and $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$ denote the safe-beam-to-helper-release-beam mapping and the closed-beam-to-helper-relay-beam mapping, respectively.

1) Critical and Helper Satellite Identification: In each time slot t , the critical satellite $s^c(t)$ is identified as the visible LEO satellite that contributes the largest EPFD risk at the GS. Since the EPFD violation is mainly caused by the critical satellite, beam shutdown is applied only to beams on $s^c(t)$, while helper and other visible satellites remain available for service continuity. The along-track north and south neighboring satellites of $s^c(t)$ are

selected as helper satellites, denoted by $s^N(t)$ and $s^S(t)$, respectively.

2) EPFD-Driven Beam Status Determination: For each beam b on the critical satellite, the beam-level EPFD risk score is defined as

$$q_{c,b}(t) = \text{EPFD}_{s^c,b}(t).$$

The critical beams are sorted in descending order of $q_{c,b}(t)$. A greedy shutdown rule is then applied: at each step, the beam with the highest EPFD risk score is shut down until the aggregate EPFD at the GS satisfies the applicable EPFD limit.

The beams shut down by this procedure form the closed-beam set $\mathcal{B}_c^{\text{off}}(t)$, and the remaining active beams form the safe-beam set:

$$\mathcal{B}_c^{\text{safe}}(t) = \mathcal{B}_{s^c} \setminus \mathcal{B}_c^{\text{off}}(t).$$

Because this beam status determination depends only on beam-level EPFD contributions and the EPFD limit, it is independent of user distribution and can be precomputed before online user reassociation.

3) Beam-Pair Mapping Construction: Given the closed-beam set $\mathcal{B}_c^{\text{off}}(t)$ and the safe-beam set $\mathcal{B}_c^{\text{safe}}(t)$, SAPR-R constructs beam-pair mappings based on footprint overlap between the critical satellite and its helper satellites.

For SAHR, the safe-beam-to-helper-release-beam mapping is defined as

$$\begin{aligned} \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t) = \{ & (b_c^{\text{safe}}, s_h, b_h^{\text{rel}}) : \\ & b_c^{\text{safe}} \in \mathcal{B}_c^{\text{safe}}(t), \\ & s_h \in \{s^N(t), s^S(t)\}, \\ & b_h^{\text{rel}} \in \mathcal{B}_{s_h}, \\ & \mathcal{F}_{s^c, b_c^{\text{safe}}}(t) \cap \mathcal{F}_{s_h, b_h^{\text{rel}}}(t) \neq \emptyset \}. \end{aligned}$$

This mapping identifies the helper beams whose users can be reassociated to critical safe beams.

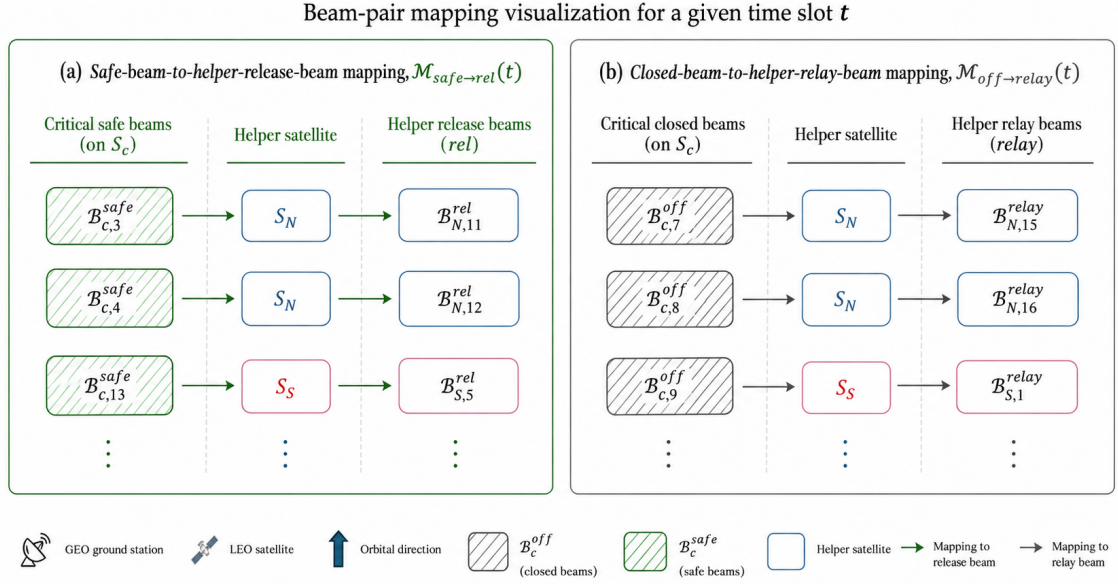


Figure 4.2: Beam-pair mapping tables at time slot t . (a) $\mathcal{M}_{safe \rightarrow rel}(t)$: critical safe beams on s^C mapped to helper release beams on s^N/s^S (SAHR). (b) $\mathcal{M}_{off \rightarrow relay}(t)$: critical closed beams mapped to helper relay beams (BPLR).

For BPLR, the closed-beam-to-helper-relay-beam mapping is defined as

$$\begin{aligned}
 \mathcal{M}_{off \rightarrow relay}(t) &= \{(b_c^{off}, s_h, b_h^{relay}) : \\
 &\quad b_c^{off} \in \mathcal{B}_c^{off}(t), \\
 &\quad s_h \in \{s^N(t), s^S(t)\}, \\
 &\quad b_h^{relay} \in \mathcal{B}_{s_h}, \\
 &\quad \mathcal{F}_{s^C, b_c^{off}}(t) \cap \mathcal{F}_{s_h, b_h^{relay}}(t) \neq \emptyset\}.
 \end{aligned}$$

This mapping identifies the helper beams that can serve users originally located under critical closed beams.

Figure 4.2 illustrates the two mappings stored in $\mathcal{R}_B(t)$: $\mathcal{M}_{safe \rightarrow rel}(t)$ for SAHR and $\mathcal{M}_{off \rightarrow relay}(t)$ for BPLR. These mappings rely only on footprint overlap and therefore can be applied to general multi-beam LEO systems, as long as the beam footprints and helper-satellite geometry are known. In the considered along-track helper scenario, the overlap is local and can be handled within predefined beam pairs, avoiding a global user-to-beam matching process.

For each helper satellite, helper beams are assigned to one of two roles in a time slot: release beams for SAHR and relay beams for BPLR. To avoid role conflict, the two sets are kept disjoint:

$$\mathcal{B}_{s_h}^{\text{rel}}(t) \cap \mathcal{B}_{s_h}^{\text{relay}}(t) = \emptyset, \quad \forall s_h \in \{s^{\text{N}}(t), s^{\text{S}}(t)\}.$$

4.3 Safe-Beam Assisted Helper Relief

After loading $\mathcal{R}_B(t)$, the online procedure starts with SAHR, as illustrated in Fig. 4.3. SAHR does not serve critical closed-beam users directly. Instead, it uses critical safe beams to take over selected users originally served by helper release beams, so that the corresponding helper satellite can make more power available for subsequent relay-beam boosting.

For each mapping tuple $(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}) \in \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$, SAPR-R identifies the helper users that can be transferred from the helper release beam to the critical safe beam. Let u_h denote a helper user. The candidate helper-user set is defined as

$$\begin{aligned} \mathcal{V}_{\text{rel}}(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}, t) = \\ \left\{ u_h \in \mathcal{U}(t) : x_{u_h, s_h, b_h^{\text{rel}}}(t) = 1, \right. \\ \left. u_h \in \mathcal{F}_{s^{\text{C}}, b_c^{\text{safe}}}(t) \right\}. \end{aligned}$$

The first condition means that user u_h is currently served by helper release beam b_h^{rel} on helper satellite s_h . The second condition means that the same user is also covered by the critical safe beam b_c^{safe} . Therefore, users in $\mathcal{V}_{\text{rel}}(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}, t)$ are eligible to be reassociated from the helper release beam to the critical safe beam.

If candidate helper user u_h is reassociated to b_c^{safe} , the reduction in the required transmit power of the helper release beam is defined as

$$\Delta P_{u_h}^{\text{rel}}(t) = P_{b_h^{\text{rel}}}^{\text{req,old}}(t) - P_{b_h^{\text{rel}}}^{\text{req,new}}(u_h, t).$$

Here, $P_{b_h^{\text{rel}}}^{\text{req,old}}(t)$ denotes the required transmit power of helper release beam b_h^{rel} before transferring u_h , while $P_{b_h^{\text{rel}}}^{\text{req,new}}(u_h, t)$ denotes the required transmit power of the same beam after u_h is removed from that beam. Therefore, $\Delta P_{u_h}^{\text{rel}}(t)$ represents the amount of helper transmit power released by transferring u_h to the critical safe beam.

Candidates are ranked in descending order of $\Delta P_{u_h}^{\text{rel}}(t)$, so users that release more helper power are considered first. A trial reassociation is accepted only when the transferred helper user and the original users on the critical safe beam do not lose satisfaction:

$$\begin{aligned} \text{sat}_{u_h}^{\text{new}}(t) &\geq \text{sat}_{u_h}^{\text{old}}(t), \\ \text{sat}_{u_s}^{\text{new}}(t) &\geq \text{sat}_{u_s}^{\text{old}}(t), \quad \forall u_s \in \mathcal{U}_{b_c^{\text{safe}}}^{\text{ori}}(t). \end{aligned}$$

Accepted transfers form the set $\mathcal{A}_{\text{rel}}(s_h, t)$. The total released power on helper satellite s_h is

$$P_{s_h}^{\text{rel}}(t) = \sum_{u_h \in \mathcal{A}_{\text{rel}}(s_h, t)} \Delta P_{u_h}^{\text{rel}}(t).$$

The available power budget for relay-beam boosting consists of the unused satellite power under the default beam-power profile and the power released by SAHR:

$$P_{s_h}^{\text{pool}}(t) = P_{s_h}^{\text{unused}}(t) + P_{s_h}^{\text{rel}}(t),$$

where

$$P_{s_h}^{\text{unused}}(t) = P_{s_h}^{\text{max}} - \sum_{b \in \mathcal{B}_{s_h}} P_{s_h, b}^{\text{def}}.$$

Helper beams that are not selected as release beams or relay beams keep their default transmit powers.

4.4 Helper Relay Beam Boosting

After SAHR, each helper satellite s_h uses its available relay power budget $P_{s_h}^{\text{pool}}(t)$, defined in the previous subsection, to increase the transmit power of helper relay beams. For helper

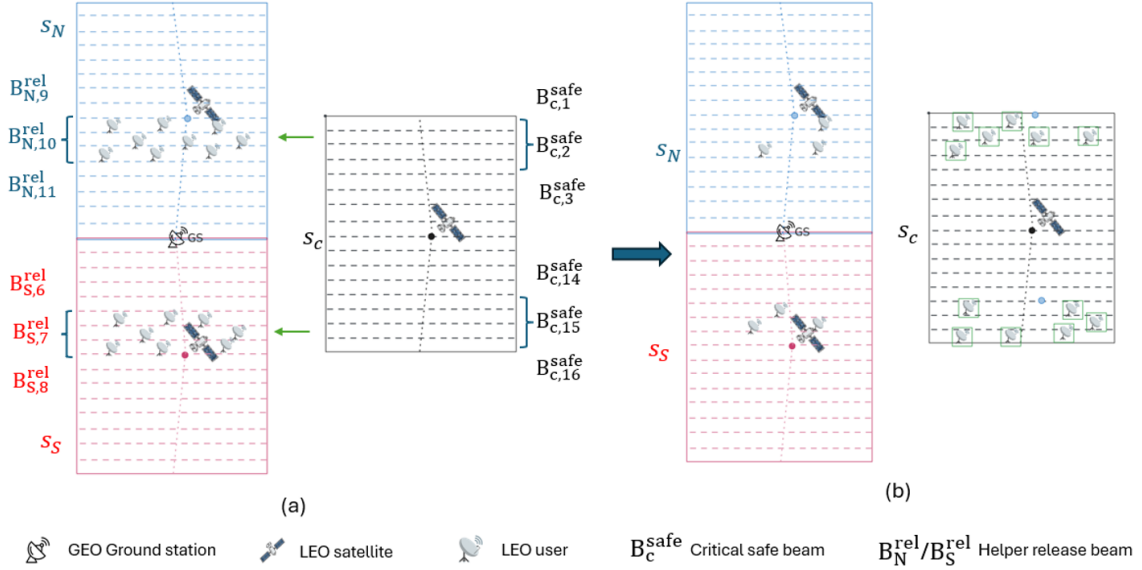


Figure 4.3: Illustration of Safe-Beam Assisted Helper Relief (SAHR). Critical safe beams take over eligible helper users from helper release beams, and the released power is returned to the helper satellite power pool.

satellite s_h , the relay beam set is defined as

$$\mathcal{B}_{s_h}^{\text{relay}}(t) = \{b_h^{\text{relay}} : \exists b_c^{\text{off}} \in \mathcal{B}_c^{\text{off}}(t) : (b_c^{\text{off}}, s_h, b_h^{\text{relay}}) \in \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)\}.$$

This set contains the helper beams on s_h that are mapped to at least one closed beam on the critical satellite and can therefore be used for BPLR.

To estimate the relay demand of each helper relay beam, SAPR-R first identifies the candidate closed-beam users that can be served by that beam. Let u_c denote a user originally served by a closed beam on the critical satellite. For helper relay beam b_h^{relay} , the candidate user set is defined as

$$\begin{aligned} \mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t) = & \{u_c \in \mathcal{U}(t) : \exists b_c^{\text{off}} \in \mathcal{B}_c^{\text{off}}(t) : \\ & (b_c^{\text{off}}, s_h, b_h^{\text{relay}}) \in \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t), \\ & x_{u_c, s_c, b_c^{\text{off}}}(t) = 1, u_c \in \mathcal{F}_{s_h, b_h^{\text{relay}}}(t)\}. \end{aligned}$$

The condition $x_{u_c, s_c, b_c^{\text{off}}}(t) = 1$ means that user u_c is originally associated with the closed

beam b_c^{off} on the critical satellite. The footprint condition means that the same user is also covered by helper relay beam b_h^{relay} . Therefore, $\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)$ represents the closed-beam users that are eligible to be relayed by b_h^{relay} .

The relay demand of helper relay beam b_h^{relay} is defined as the number of its candidate users:

$$D_{b_h^{\text{relay}}}^{\text{relay}}(t) = \left| \mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t) \right|.$$

This demand value is used only to prioritize relay-beam boosting. The actual user reassociation is performed later in BPLR after the final transmit power of each helper relay beam is determined.

For a helper relay beam b_h^{relay} , let $\Delta P_{b_h^{\text{relay}}}^{\text{boost,max}}(t)$ denote the additional power required to increase its transmit power from the default value to the beam-level maximum:

$$\Delta P_{b_h^{\text{relay}}}^{\text{boost,max}}(t) = P_{s_h, b_h^{\text{relay}}}^{\text{max}} - P_{s_h, b_h^{\text{relay}}}^{\text{def}}.$$

If $P_{s_h}^{\text{pool}}(t)$ is sufficient to raise all helper relay beams to their beam-level maximum powers, all relay beams are fully boosted. Otherwise, helper relay beams are sorted in descending order of $D_{b_h^{\text{relay}}}^{\text{relay}}(t)$, and beams with higher relay demand receive additional power first.

For each helper relay beam, if the remaining available relay power budget is sufficient, its transmit power is increased to $P_{s_h, b_h^{\text{relay}}}^{\text{max}}$. If the remaining budget is positive but insufficient for full boosting, the remaining power is assigned as a partial boost above $P_{s_h, b_h^{\text{relay}}}^{\text{def}}$. This avoids leaving unused relay power and increases the available capacity for BPLR.

After this step, each helper relay beam proceeds to BPLR with its final transmit power.

4.5 Beam-Pair Local Relay

After relay-beam boosting, BPLR performs the actual reassociation of closed-beam users to helper relay beams, as illustrated in Fig. 4.4. For each mapping tuple $(b_c^{\text{off}}, s_h, b_h^{\text{relay}}) \in \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$, BPLR considers the users in $\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)$ that were originally served by the closed beam b_c^{off} . Since $\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)$ already includes the coverage condition of the

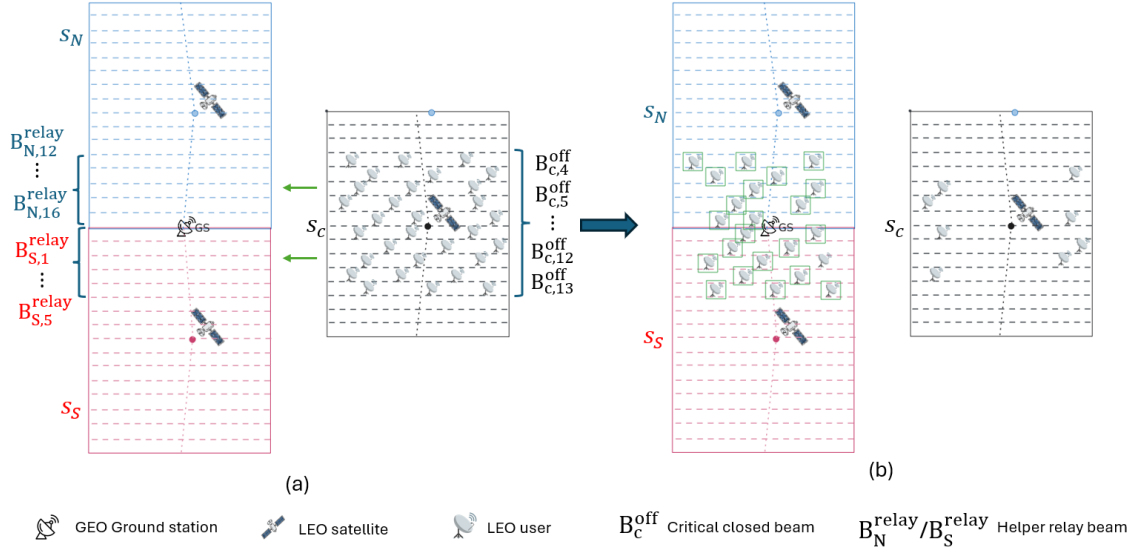


Figure 4.4: Illustration of Beam-Pair Local Relay (BPLR). Candidate critical users in closed beams are reassociated to the corresponding helper relay beams according to $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$.

helper relay beam, these users are eligible to be reassociated to b_h^{relay} .

Let u_c denote a closed-beam user considered in this beam pair. For each candidate user u_c , let $R_{u_c, b_h^{\text{relay}}}(t)$ denote the achievable service rate if u_c is served by helper relay beam b_h^{relay} using the final transmit power obtained after relay-beam boosting. Candidates are ranked in descending order of $R_{u_c, b_h^{\text{relay}}}(t)$, so users with better relay link quality are considered first.

Let $C_{b_h^{\text{relay}}}^{\text{rem}}(t)$ denote the remaining relay capacity of helper relay beam b_h^{relay} after preserving the service of its original helper users. This remaining capacity is allocated sequentially to the sorted closed-beam users. If $C_{b_h^{\text{relay}}}^{\text{rem}}(t)$ can satisfy the demand $D_{u_c}(t)$ of user u_c , the user is reassociated to b_h^{relay} and its satisfaction is updated. If the remaining capacity is positive but insufficient to fully satisfy the next user, partial capacity is assigned to that user and the beam-pair process stops. When the remaining capacity is exhausted, no further users are reassociated in this beam pair.

All relay decisions follow the single-association constraint. A relayed user is moved from the critical closed beam to the helper relay beam and is served by only one beam in the time slot.

4.6 Overall Algorithm

The online procedure uses a single time-slot-based algorithm. In each time slot, SAPR-R first runs SAHR and relay-beam boosting, then runs BPLR using the updated helper relay-beam powers. The complete procedure is given in Algorithm 1.

Algorithm 1 SAPR-R Online Operation

Input: $\{\mathcal{R}_B(t)\}_{t=1}^T$, initial $\mathbf{x}(t)$, $\mathcal{U}(t)$, $D_u(t)$

Input: $P_{s,b}^{\text{def}}$, $P_{s,b}^{\text{max}}$, P_s^{max} , $\mathcal{F}_{s,b}(t)$, capacity/satisfaction model

Output: $\mathbf{x}^{\text{final}}(t)$, $P_{s,b}^{\text{final}}(t)$, $\text{sat}_u^{\text{final}}(t)$

- 1: **for** $t = 1$ **to** T **do**
- 2: Load $\mathcal{R}_B(t)$; initialize beam powers per default profile and closed/safe beam status
- 3: Initialize $P_{s_h}^{\text{rel}}(t) = 0$ for each $s_h \in \{s^N(t), s^S(t)\}$
- 4: **Stage 1: SAHR**
- 5: **for all** $(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}) \in \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$ **do**
- 6: Build $\mathcal{V}_{\text{rel}}(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}, t)$ using $\mathbf{x}(t)$ and beam footprints
- 7: Compute $\Delta P_{u_h}^{\text{rel}}(t)$ for each $u_h \in \mathcal{V}_{\text{rel}}(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}, t)$
- 8: Sort candidate helper users by $\Delta P_{u_h}^{\text{rel}}(t)$ descending
- 9: **for all** u_h in sorted order **do**
- 10: Trial reassociation of u_h from b_h^{rel} to b_c^{safe}
- 11: **if** transferred helper user and original users on b_c^{safe} do not lose satisfaction **then**
- 12: Accept the reassociation
- 13: Update $\mathbf{x}(t)$, satisfaction values, and $P_{s_h}^{\text{rel}}(t)$
- 14: **end if**
- 15: **end for**
- 16: **end for**
- 17: **Stage 1 continued: helper relay beam boosting**
- 18: **for all** $s_h \in \{s^N(t), s^S(t)\}$ **do**
- 19: Compute unused power $P_{s_h}^{\text{unused}}(t)$ under default beam-power profile
- 20: Compute relay power budget $P_{s_h}^{\text{pool}}(t) = P_{s_h}^{\text{unused}}(t) + P_{s_h}^{\text{rel}}(t)$
- 21: Extract helper relay beam set $\mathcal{B}_{s_h}^{\text{relay}}(t)$ from $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$
- 22: **for all** $b_h^{\text{relay}} \in \mathcal{B}_{s_h}^{\text{relay}}(t)$ **do**
- 23: Build $\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)$
- 24: Compute relay demand $D_{b_h^{\text{relay}}}^{\text{relay}}(t) = |\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)|$
- 25: Compute $\Delta P_{b_h^{\text{relay}}}^{\text{boost,max}}(t)$ for boosting b_h^{relay} to beam-level maximum
- 26: **end for**
- 27: Sort helper relay beams by $D_{b_h^{\text{relay}}}^{\text{relay}}(t)$ descending
- 28: **for all** b_h^{relay} in sorted order **do**
- 29: Allocate $P_{s_h}^{\text{pool}}(t)$ to b_h^{relay} : full boost, partial boost, or default power
- 30: Update final transmit power of b_h^{relay}
- 31: **end for**
- 32: **end for**
- 33: **Stage 2: Beam-Pair Local Relay**
- 34: **for all** $(b_c^{\text{off}}, s_h, b_h^{\text{relay}}) \in \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$ **do**
- 35: Select from $\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)$ users originally served by b_c^{off}
- 36: Compute remaining relay capacity of b_h^{relay} after preserving original helper users
- 37: Compute $R_{u_c, b_h^{\text{relay}}}(t)$ for each candidate u_c using final power of b_h^{relay}
- 38: Sort candidate closed-beam users by $R_{u_c, b_h^{\text{relay}}}(t)$ descending
- 39: **for all** u_c in sorted order **do**
- 40: **if** remaining relay capacity can satisfy $D_{u_c}(t)$ **then**
- 41: Reassociate u_c to b_h^{relay}
- 42: Update $\text{sat}_{u_c}(t)$ and remaining relay capacity
- 43: **else if** remaining relay capacity > 0 **then**
- 44: Assign partial capacity to u_c
- 45: Update $\text{sat}_{u_c}(t)$; set remaining relay capacity to 0; **break**
- 46: **else**
- 47: **break**
- 48: **end if**
- 49: **end for**
- 50: **end for**
- 51: Validate single-association, beam power, satellite total power, and EPFD constraints
- 52: Store $\mathbf{x}^{\text{final}}(t)$, $P_{s,b}^{\text{final}}(t)$, and $\text{sat}_u^{\text{final}}(t)$
- 53: **end for**

4.7 Complexity Discussion

SAPR-R avoids directly solving the mixed-integer nonlinear optimization problem formulated in Chapter III. Since the beam relation record $\mathcal{R}_B(t)$ is precomputed, the online procedure does not need to re-identify the critical satellite, helper satellites, or beam-pair mappings in each time slot. The online computation is therefore limited to candidate construction, local sorting, power allocation, and sequential reassociation within predefined beam pairs.

Let K_t^{SAHR} denote the number of candidate helper users considered in SAHR at time slot t , and let K_t^{BPLR} denote the number of candidate closed-beam users considered in BPLR. In the SAHR stage, candidate construction and released-power computation are linear in K_t^{SAHR} , while sorting the candidates by released power dominates the complexity:

$$\mathcal{O}(K_t^{\text{SAHR}} \log K_t^{\text{SAHR}}).$$

In the relay-beam boosting stage, let H_t denote the number of helper relay beams in time slot t . Sorting helper relay beams by relay demand requires

$$\mathcal{O}(H_t \log H_t).$$

In the BPLR stage, candidate users are sorted by achievable relay service rate, and then allocated sequentially according to the remaining relay capacity. The dominant complexity is therefore

$$\mathcal{O}(K_t^{\text{BPLR}} \log K_t^{\text{BPLR}}).$$

Over all time slots, the online complexity is

$$\mathcal{O}\left(\sum_{t=1}^T (K_t^{\text{SAHR}} \log K_t^{\text{SAHR}} + H_t \log H_t + K_t^{\text{BPLR}} \log K_t^{\text{BPLR}})\right).$$

Thus, SAPR-R reduces the online decision process to local sorting and sequential allo-

cation within precomputed beam relations, instead of solving a global association and continuous power-control problem in every time slot.

4.8 Summary

This chapter presented SAPR-R for maintaining user service under the GS EPFD constraint during critical time slots. The method first precomputes time-slot beam relation records from predictable satellite geometry, beam footprints, and beam-level EPFD information. During online operation, SAPR-R applies SAHR, helper relay beam boosting, and BPLR in sequence.

The staged design separates the original coupled problem into local decisions. SAHR uses critical safe beams to take over selected helper users and release helper satellite power. Helper relay beam boosting allocates the available relay power budget according to relay demand. BPLR then reassociates closed-beam users to feasible helper relay beams within predefined beam pairs. In this way, SAPR-R improves user satisfaction during critical periods while satisfying the EPFD constraint.

Chapter 5

Evaluation

5.1 Evaluation Overview

This chapter evaluates the performance of the proposed Safe-Beam Assisted Power-Release Relay (SAPR-R) method, focusing on its ability to improve user satisfaction and service continuity under the aggregate EPFD constraint. The evaluation consists of three main parts. First, we observe the aggregate EPFD variation during the critical satellite pass over the GS, as well as the number of beams that must be shut down to satisfy the EPFD constraint. This analysis confirms the existence of EPFD violation and beam shutdown demand during the critical pass, and further supports the rationale of designing a relay-based recovery method. Second, we compare different interference mitigation methods in terms of service performance and relay capability. Finally, we further analyze the impact of different EPFD limits on user satisfaction to evaluate the stability and applicability of SAPR-R under different interference constraint conditions.

5.2 Simulation Setup

To evaluate the performance of the proposed SAPR-R method under the NGSO–GSO co-existence scenario, we design a simulation framework that combines STK and MATLAB. This framework is used to construct the LEO constellation, obtain the time-varying satellite geometry, and further evaluate the performance of different interference mitigation

Table 5.1: Simulation parameters.

Parameter	Value
LEO constellation	
Altitude (km)	1200
Orbit plane inclination	87.9°
Time slot	1 s
Maximum transmit power	16.8 W per satellite
Maximum antenna gain	34.33 dBi
Downlink frequency	11.7 GHz
Bandwidth	250 MHz
Antenna pattern	[14] $A \exp(\beta\phi)$
Maximum tilt angle	10°
Approximated antenna gain coefficient, (A)	4.0×10^3
Approximated antenna gain coefficient, (B)	-0.0671
GSO satellite	
Altitude (km)	35785.4
Longitude	30.6
Latitude	0
Downlink frequency	11.7 GHz
GSO ground station	
Longitude	30.6
Latitude	0
Minimum elevation angle	10°
Maximum receive gain	38.1 dBi
Noise temperature	240 K
Antenna pattern	[8]
LEO user ground station	
Maximum receive gain	40.95 dBi
Noise temperature	240 K
Antenna pattern	[8]
User demand	50 Mbps
EPFD	
EPFD limit (dB(W/m ² /1 MHz))	-173.4 [5]
Reference antenna diameter	60 cm

methods in terms of EPFD compliance, downlink capacity, and user satisfaction.

The simulation uses the following two main tools:

- **STK (Satellite Tool Kit):** [20] STK is used to construct the OneWeb-like LEO constellation and generate the relative positions, visibility relationships, and beam coverage information among the LEO satellites, the GS, and users at each time slot. Based on the system information generated by STK, the dynamic geometric variation during the critical satellite pass near the GS can be simulated.
- **MATLAB:** [21] MATLAB is used to process the system information generated by STK and perform offline preprocessing. This includes constructing the beam-user association, critical/helper satellite relationship, and critical-beam/helper-beam mappings. MATLAB is then used to execute the proposed SAPR-R method and calculate the EPFD

and user satisfaction performance of each method at different time slots.

The main simulation parameters used in this chapter are summarized in Table 5.1. The table integrates the settings related to the LEO constellation, GSO satellite, GS, LEO user ground station, and EPFD evaluation. These parameters are used in the following EPFD evaluation and user satisfaction comparison.

5.3 Compared Schemes

To evaluate the proposed SAPR-R method, we compare it with several interference mitigation methods. All methods are evaluated under the same simulation parameters to ensure a fair comparison. The compared schemes are described as follows:

- **Beam shutdown only:** When the aggregate EPFD exceeds the limit, the high-interference beams on the critical satellite are directly shut down to satisfy the EPFD constraint. This method does not perform relay.
- **PC + Tilt:** This method represents a power-control and satellite-tilting based interference mitigation strategy. It reduces the interference toward the GS by adjusting the transmit power and satellite tilt angle. Although this method can help satisfy the EPFD constraint, it does not provide a relay-based recovery mechanism for users.
- **Only BPLR:** This method only applies the BPLR stage. Helper relay beams serve users located under the critical closed beams using the default power of the helper satellite. The SAHR stage is not applied.
- **SAPR-R (Proposed):** This method includes both the SAHR and BPLR stages.

5.4 Evaluation Results

Based on the simulation setup and compared schemes described above, this section evaluates the performance of SAPR-R during the critical pass. The analysis is divided into four parts. First, the aggregate EPFD and the number of closed beams are examined to verify whether SAPR-R can still satisfy the EPFD constraint after relay-based recovery, and to observe the beam shutdown behavior of the critical satellite at different time slots.

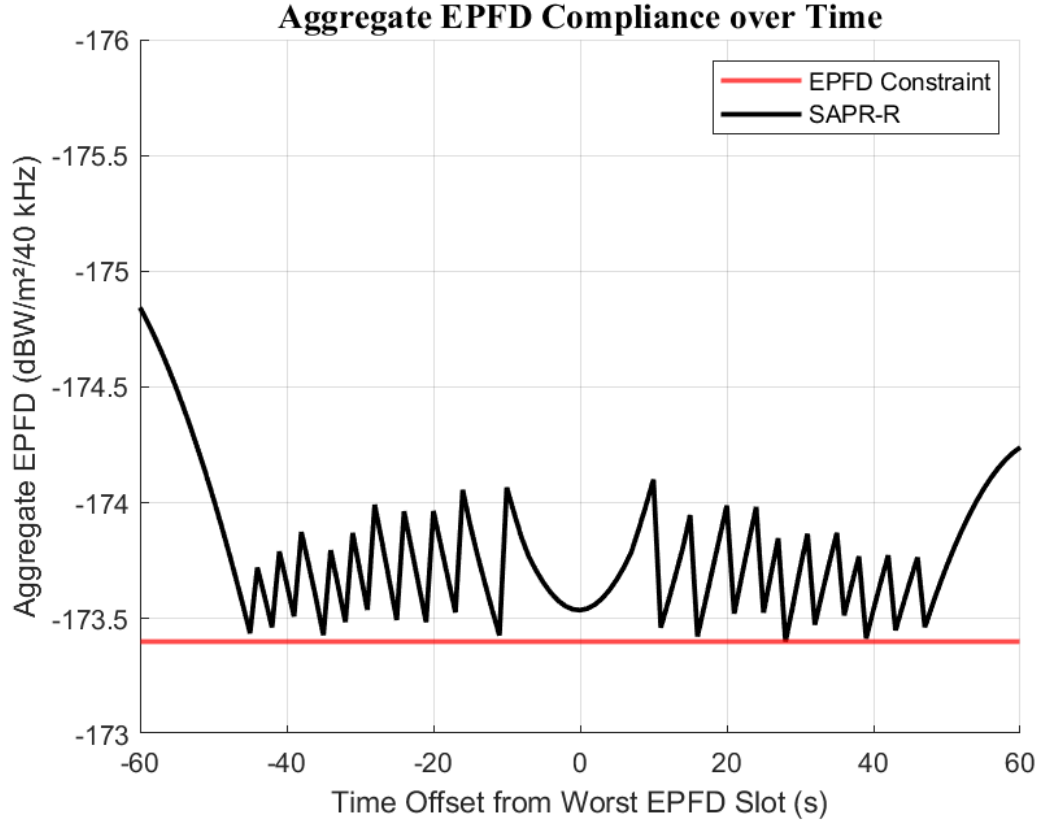


Figure 5.1: Aggregate EPFD over time during the critical pass.

Second, the average satisfaction of different methods is compared under different user loads to analyze how increasing system load affects service performance. Third, the number of relayed critical users is used to explain the main reason why SAPR-R improves satisfaction, and the satisfaction CDF is further used to observe the satisfaction distribution of all users. Finally, the average user satisfaction of SAPR-R under different EPFD limits is evaluated to observe how the strictness of the interference limit affects service performance.

1) EPFD Compliance and Beam Shutdown:

This subsection first analyzes the aggregate EPFD after applying SAPR-R and further observes the number of beams that must be shut down on the critical satellite to satisfy the EPFD constraint. This analysis is used to verify whether the proposed method can still satisfy the EPFD limit after relay-based recovery, and to show the beam shutdown behavior during the critical pass.

Fig. 5.1 shows the aggregate EPFD variation during the critical satellite pass near

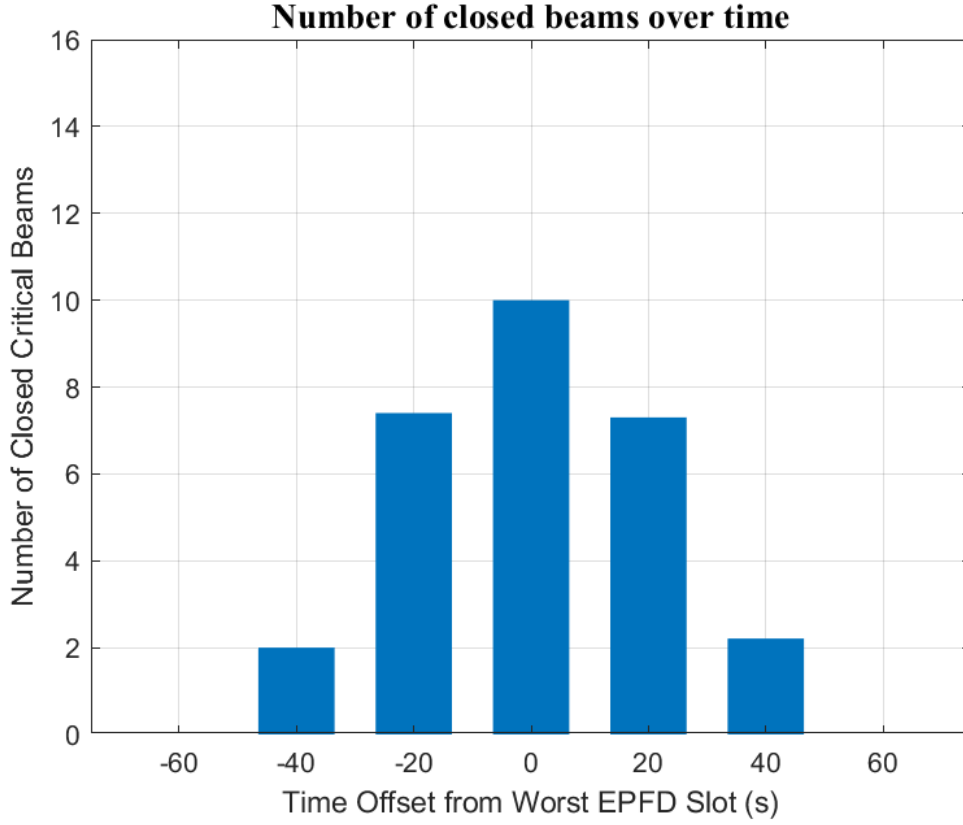


Figure 5.2: Number of closed beams over time during the critical pass.

the GS. The red line represents the EPFD constraint, and the black line represents the aggregate EPFD after applying SAPR-R. As shown in the figure, SAPR-R keeps the aggregate EPFD below the limit during the entire observation period. This indicates that the proposed method does not cause EPFD violation while performing service recovery for closed-beam users. Therefore, SAPR-R can compensate for the service loss caused by beam shutdown while maintaining EPFD compliance.

Fig. 5.2 shows the number of beams that must be shut down on the critical satellite at different time offsets. When the satellite approaches the worst EPFD slot, the number of closed beams gradually increases and reaches its maximum around the worst EPFD slot. After the satellite moves away from GS, the number of closed beams gradually decreases. This result shows that the beam shutdown demand changes with the satellite–GS geometry.

2) User Satisfaction under Different User Loads:

This subsection compares the average satisfaction of different interference mitigation methods under different user loads. By changing the number of users, we can observe

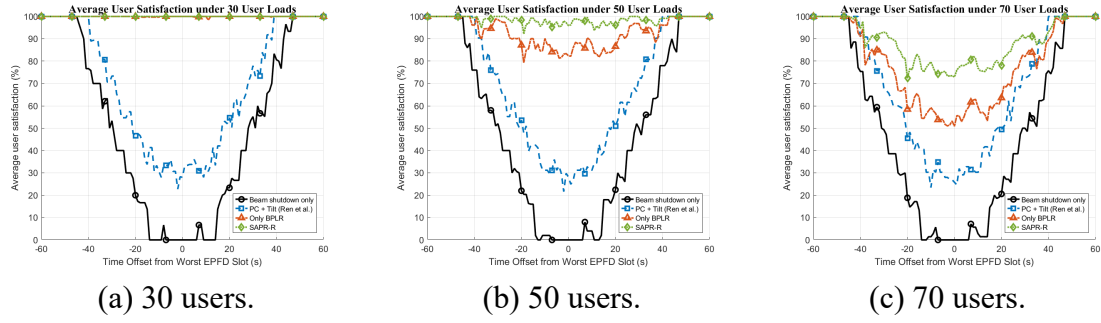


Figure 5.3: Average satisfaction under different user loads. (a) 30 users. (b) 50 users. (c) 70 users.

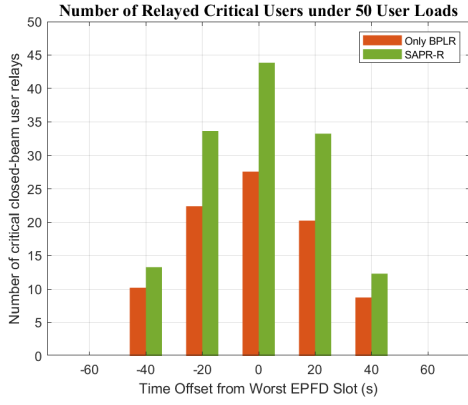
how the service performance changes as the system load increases, and evaluate whether SAPR-R can maintain user satisfaction under different load conditions.

Fig. 5.3 shows the average satisfaction under three user load scenarios: 30 users, 50 users, and 70 users. As shown in the figure, Beam shutdown only has the lowest satisfaction because it does not provide relay service for users affected by closed beams. PC + Tilt can reduce the interference toward the GS, but it does not recover the service of users affected by beam shutdown. Only BPLR can serve part of the closed-beam users through helper relay beams, so it achieves better satisfaction than Beam shutdown only and PC + Tilt.

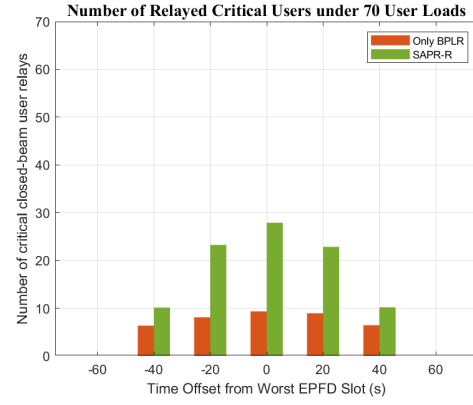
Compared with the other methods, SAPR-R maintains higher average satisfaction under all three user loads. When the user load is low, the helper satellite usually has enough available power, so the difference between Only BPLR and SAPR-R is small. However, as the user load increases, more closed-beam users need relay service, and Only BPLR becomes limited by the default power of the helper satellite. In contrast, SAPR-R uses SAHR to release available power from the helper satellite and improve the relay capability of helper relay beams. Therefore, SAPR-R can maintain more stable service performance under medium and high user loads.

3) Relay Capability and Satisfaction Distribution:

This subsection further analyzes why SAPR-R improves user satisfaction and observes the satisfaction distribution of different methods. Since 30 users is a low-load scenario, the default available power of the helper satellite is usually sufficient to support most relay demand. Therefore, the difference in relay capability is not obvious under 30 users. To

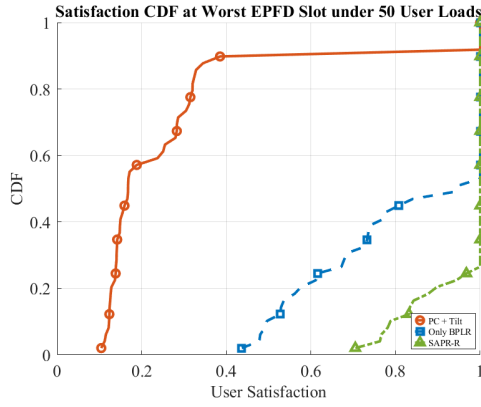


(a) 50 users.

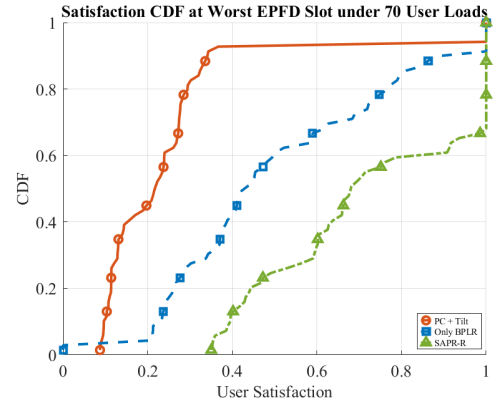


(b) 70 users.

Figure 5.4: Number of relayed critical users with and without SAHR. (a) 50 users. (b) 70 users.



(a) 50 users.



(b) 70 users.

Figure 5.5: CDF of user satisfaction under different interference mitigation methods.

better show the effect of the proposed algorithm, this subsection focuses on the 50-user and 70-user scenarios.

Fig. 5.4 compares the number of successfully relayed critical users between Only BPLR and SAPR-R under the 50-user and 70-user scenarios. As shown in the figure, SAPR-R serves more critical users in both load scenarios. The difference becomes more obvious around the worst EPFD slot, where the beam shutdown demand is the most severe. This result shows the necessity of Stage 1 in SAPR-R.

Fig. 5.5 shows the satisfaction CDF of three interference mitigation methods. This figure focuses on the user-level satisfaction distribution. A CDF curve that shifts further to the right means that more users maintain higher satisfaction. In contrast, if the curve rises quickly in the low-satisfaction region, more users suffer from service degradation.

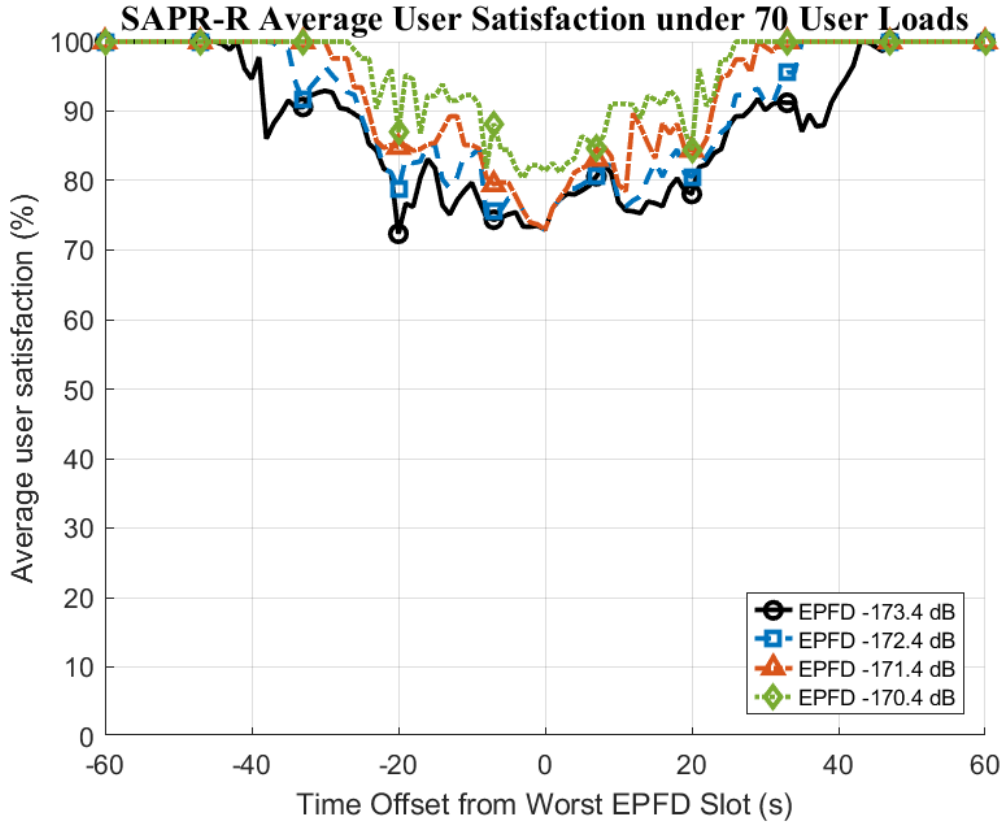


Figure 5.6: Average user satisfaction of SAPR-R under different EPFD limits.

As shown in the figure, the CDF curve of SAPR-R shifts more toward the high-satisfaction region than the other methods. This means that more users can maintain better service quality. The result shows that SAPR-R not only improves average satisfaction, but also reduces the proportion of low-satisfaction users, making the overall user service quality more stable.

4) Impact of Different EPFD Limits:

This subsection further evaluates the impact of different EPFD limits on the service performance of SAPR-R. When the EPFD limit becomes stricter, the allowed aggregate interference becomes lower. As a result, the critical satellite may need to shut down more beams to satisfy the constraint, and the relay demand of closed-beam users may also increase. This analysis is used to observe whether SAPR-R can maintain stable average user satisfaction under different EPFD limits.

Fig. 5.6 shows the average user satisfaction of SAPR-R under different EPFD limits. As shown in the figure, when the EPFD limit is more relaxed, SAPR-R can maintain

higher average user satisfaction. This result shows that the strictness of the EPFD limit directly affects user satisfaction. It also indicates that SAPR-R can achieve better service performance under more relaxed interference constraints.

Overall, SAPR-R keeps the aggregate EPFD below the limit during the critical pass and uses SAHR to improve the ability of helper relay beams to serve critical users. Compared with other interference mitigation methods, SAPR-R maintains higher average satisfaction under different user loads and reduces the proportion of low-satisfaction users. In addition, the results under different EPFD limits show that the strictness of the interference constraint affects user satisfaction, and SAPR-R can maintain better service performance under more relaxed constraints. Overall, SAPR-R achieves a better balance between EPFD compliance and service continuity.

Chapter 6

Conclusion

This thesis proposed Safe-Beam Assisted Power-Release Relay (SAPR-R), a staged relay-based service recovery framework for EPFD-constrained multi-beam LEO satellite systems. The proposed method exploits predictable satellite motion, beam footprints, and coverage overlap among neighboring LEO satellites. By precomputing the critical satellite, helper satellites, closed beams, safe beams, and beam-pair mappings for each time slot, SAPR-R reduces the online decision complexity and enables local relay-based reassociation during critical periods.

SAPR-R consists of three main stages. First, Safe-Beam Assisted Helper Relief (SAHR) allows critical safe beams to take over selected helper users without reducing user satisfaction. This process releases available power from the helper satellite. Second, the released power is allocated to helper relay beams according to relay demand, improving their ability to serve users affected by closed-beam shutdown. Third, Beam-Pair Local Relay (BPLR) reassociates closed-beam users to feasible helper relay beams within predefined beam pairs. Through this staged design, SAPR-R improves the relay capability of helper satellites while maintaining the single-association, beam power, satellite power budget, and GS EPFD constraints.

The simulation results show that SAPR-R keeps the aggregate EPFD below the required limit during the critical satellite pass. Compared with beam shutdown only, PC + Tilt, and Only BPLR, SAPR-R achieves higher average user satisfaction under different user loads. The results also show that SAHR increases the number of successfully relayed

critical users, especially around the worst EPFD slot where the relay demand is the highest. In addition, the satisfaction CDF results indicate that SAPR-R reduces the proportion of low-satisfaction users and provides more stable service quality. The evaluation under different EPFD limits further shows that user satisfaction is affected by the strictness of the interference constraint, while SAPR-R can achieve better service performance when the constraint is more relaxed.

Overall, SAPR-R provides a practical service recovery approach for multi-beam LEO satellite systems under NGSO–GSO coexistence. The proposed method does not rely on solving a high-complexity global optimization problem in every time slot. Instead, it uses precomputed beam relations, predictable beam overlap, and neighboring helper satellites to support relay-based reassociation for users affected by critical-beam shutdown. Therefore, SAPR-R achieves a better balance between EPFD compliance, service continuity, and implementation complexity.

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